# Measurement Map

Companion to the manuscript skeleton, 10 October 2026

## 1 Scope

The map takes each object of the model in `MANUSCRIPT.tex` to its counterpart in the Basel capital framework [Basel Committee on Banking Supervision, 2011], the Basel liquidity standard [Basel Committee on Banking Supervision, 2013] and the Call Report instructions for regulatory capital [Federal Financial Institutions Examination Council, 2026], and says how it is observed. The FDIC panel fields are defined in the FDIC variable workbook [Federal Deposit Insurance Corporation, 2026]. The regulatory cap is that of Bayraktar et al. [2026] (BCVW). Quotation and reading IDs refer to the `lit/` records, where each quotation is checked verbatim on its page. The rearrangements of the regulatory ratios are proved in `lean/mathlib/MeasurementMap.lean`.

A status of Rule means a regulation sets the value. Report means a regulatory report records the object for each bank. Calibration means the model takes a value with no regulatory counterpart. Unobserved means neither a rule nor a report gives it. Output means the model produces it and its observation is part of the empirical design.

## 2 Terminology

One plain-English name for each object of the model. The manuscript and this map use these names and no others. Agreed means fixed with the author on 10, 13, 19 or 23 October 2026 (`notes/writing_discipline.md`, section 6). In use means the manuscript already uses the name consistently. Proposed means a plain name offered to the author and not yet agreed. The names in the last column are retired, and neither document uses them for these objects.

Table 1: Terminology.

| Name | Symbol | Meaning | Status | Replaces |
|---|---|---|---|---|
| bank | none | The institution that files the report, the agent whose policy the model solves. | Agreed | none |
| investors | none | Those who buy the bank's stock or receive its dividends, its parent holding company among them. | Agreed | none |
| insured creditors | none | Those whose claims on the bank deposit insurance covers, so that they lose nothing if it fails. | Agreed | none |
| uninsured creditors | none | Those whose claims on the bank no insurance covers: lenders of wholesale funds and holders of deposits above the insurance limit. | Agreed | none |
| supervisor | none | The bank's primary federal regulator, which supervises it and receives its reports. | Agreed | none |
| insurer | none | The deposit insurer, which insures the insured creditors' deposits. | Agreed | none |
| assets-to-liabilities ratio | $x$ | Total assets over total liabilities, the state of the model. | Agreed | none |
| equity | $X - L$ | Assets less liabilities. | In use | none |
| required capital | $Q$, $q$ | The capital regulation requires, in dollars or as a ratio to liabilities. | Agreed | none |

| Name | Symbol | Meaning | Status | Replaces |
|---|---|---|---|---|
| capital buffer | $z$, $z_t$ | The log of equity over required capital. Zero when equity equals required capital, falling without limit towards the distress boundary. | Agreed | log coordinate |
| distress boundary | $x = 1$ | The state at which equity is zero. | In use | deep in deficit |
| deficit | $z < 0$ | Equity below required capital. | In use | none |
| surplus | $z > 0$ | Equity above required capital. | In use | none |
| risky exposure | $\pi$ | The share of assets held in the risky asset. | Agreed | none |
| regulatory cap | $u(x)$ | The largest risky exposure regulation allows at each state, the smaller of its two branches. | In use | none |
| solvency branch | $(x-1)/a_1$ | The branch of the regulatory cap set by capital. It binds near the distress boundary. | In use | none |
| liquidity branch | $(x-a_2)/a_3$ | The branch of the regulatory cap set by liquidity. It binds deep in surplus. | In use | none |
| crossing point | $\bar{x}$ | The state at which the two branches are equal. | In use | none |
| cap geometry | $a_1, a_2, a_3$ | The three parameters of the regulatory cap, taken together. Never a preference. | Agreed | none |
| solvency floor | $a_1$ | The floor on equity over risky assets, the slope of the solvency branch. | Agreed | none |
| run-off rate | $a_2$ | Net 30-day outflows as a share of liabilities, the intercept of the liquidity branch. | Agreed | none |
| liquidity haircut | $a_3$ | The haircut on the risky asset in high-quality liquid assets, the slope of the liquidity branch. | Agreed | none |
| value function | $V$ | The value of the bank under its best policy, per unit of liabilities. | Agreed | none |
| asset volatility | $\sigma$ | The volatility of the risky asset. | Agreed | none |
| expected asset return | $\mu_s$ | The drift of the risky asset. | Agreed | none |
| riskless rate | $r$ | The return on the riskless asset. | In use | none |
| liability volatility | $\sigma_L$ | The volatility of the shock to liabilities. | Agreed | none |
| return-liability correlation | $c$ | The correlation between the risky return and the liability shock. | Agreed | none |
| net income on liabilities | $\gamma$ | Income earned on liabilities, net of their cost. | In use | none |
| liability drift | $\mu_L$ | The drift of liabilities, $\gamma + r_L$. | In use | none |
| deposit rate | $r_L$ | The rate paid on liabilities. | In use | none |
| discount rate | $\rho$ | The shareholders' discount rate. | Agreed | none |
| liability-net discount rate | $\rho_L$ | The discount rate of the problem written per unit of liabilities. | Agreed | none |
| return on equity | $r_E$ | Net income over equity. | Agreed | none |
| dividends | $D$ | Payments to shareholders, the singular control. | Agreed | none |
| dividend barrier | $y^*$ | The state at which dividends are paid. | Agreed | payout barrier |
| recapitalisation | $I$ | Capital raised in one issue, the impulse control. | Agreed | none |
| recapitalisation trigger | $x_L$ | The state at which capital is raised. | Agreed | injection trigger, recapitalisation boundary |
| injection target | $y_{\text{post}}$ | The state an issue reaches. | Agreed | none |
| thresholds | none | The dividend barrier and the recapitalisation trigger, taken together. | Proposed | none |
| fixed issuance cost | $K$ | The cost of an issue that does not scale with its size, per unit of liabilities. | Agreed | none |

| Name | Symbol | Meaning | Status | Replaces |
|---|---|---|---|---|
| proportional issuance cost | $\kappa$ | The cost of an issue per unit raised. | Agreed | none |
| asymmetry parameter | $\lambda_S$ | How much more the objective weighs a shortfall below the reference than a surplus above it. | Agreed | loss aversion |
| shortfall penalty | $\Lambda$ | The term of the objective that charges a shortfall below the reference. Zero when the asymmetry parameter is one. | In use | none |
| reference | $R$ | The level below which the shortfall penalty applies. | In use | none |
| expected discounted shortfall | $W$ | The expected discounted shortfall below the reference under the optimal policy. | In use | none |
| discounted injection count | $N$ | The expected discounted number of issues under the optimal policy. | In use | none |
| impulse operator | $\mathscr{M}V$ | The value of issuing at once to the best target, net of both issuance costs. | In use | none |
| buffer variance | $\zeta$ | The variance per unit of time of the capital buffer. | Agreed | z-volatility, capital buffer's volatility |
| deficit limit | $\nu_1^2$ | The limit of buffer variance at the distress boundary. | Agreed | saturation limits |
| surplus limit | $\nu_3^2$ | The limit of buffer variance deep in surplus. | Agreed | saturation limits |
| variance ratio | $\lambda_V^4$, $\nu_1^2/\nu_3^2$ | The deficit limit over the surplus limit. | Agreed | volatility ratio |
| induced variance asymmetry | $\lambda_V$ | The asymmetry read off the variance ratio, distinct from the asymmetry parameter. | Agreed | none |
| saturation scale | $\theta$ | How sharply the variance of the discrete-time recursion moves between its deficit and surplus levels. | In use | none |
| growth rate of required capital | $g$ | The growth rate of required capital. | In use | none |
| retention ratio | $b$ | The share of net income retained. | In use | none |
| surplus payout rate | $\delta_p$ | The share of surplus paid out in a period. | In use | none |
| deficit persistence | $\phi^-$ | How much of a deficit remains after one period. | In use | none |
| surplus persistence | $\phi^+$ | How much of a surplus remains after one period. | In use | none |

# 3 Map

Table 2: Model objects and their observation.

| ID | Object | Role in the model | Counterpart and observation | Status | Evidence |
|---|---|---|---|---|---|
| G1 | $x$ | The state, assets over liabilities. | Total assets over total liabilities, `ASSET/LIAB` in the FDIC panel. From the reported leverage ratio $\ell$ (Tier 1 over total assets for the leverage ratio), $x = 1/(1-\ell)$ holds only if Tier 1 equals book equity and the denominator equals total assets. | Report | `FDIC-Q1,FDIC-Q2,RCR-Q4,RCR-D3, MeasurementMap. leverage_of_state` |
| G2 | $\pi$ | The share of assets in the risky asset. | Risk-weighted assets over total assets, `RWAJT/ASSET`, under risk weights of 0 on the riskless and 1 on the risky asset. With a positive weight on the riskless asset the identification fails. | Report | `FDIC-Q3,RCR-Q5,RCR-D2,MeasurementMap. rwa_two_assets, MeasurementMap. control_rwa_needs_ zero_weight_on_ riskless` |

| ID | Object | Role in the model | Counterpart and observation | Status | Evidence |
|---|---|---|---|---|---|
| G3 | $X - L$ | Equity, assets less liabilities. | CET1 capital or Tier 1 capital (RBCT1J) as reported. Both are net of regulatory adjustments, so they equal book equity only when the adjustments vanish. | Report | FDIC-Q4,B3-Q3,RCR-Q1, RCR-Q3,B3-D2 |
| G4 | $a_1$ | The solvency floor, the floor on equity over risky assets, the solvency branch $(x-1)/a_1$ of the cap. | The CET1 minimum of 4.5% of risk-weighted assets, which is BCVW's baseline 0.045, or the Tier 1 minimum of 6.0%. The bank's own ratio is reported as CET1 (RBCT1CER) or Tier 1 capital over risk-weighted assets. Banks electing the community bank leverage ratio report neither. | Rule | FDIC-Q5,FDIC-Q6,BCVW-Q9,BCVW-Q13,B3-Q1,B3-Q2,B3-D1,RCR-Q2,RCR-Q6,RCR-Q7,RCR-D1, MeasurementMap. solvency_iff, MeasurementMap. control_solvency_ needs_positive_ exposure |
| G5 | $a_2$ | Net 30-day outflows as a share of liabilities, in the liquidity branch $(x-a_2)/a_3$ of the cap. | A run-off rate on all liabilities. BCVW's baseline 0.05 is the factor for stable insured retail deposits, and less stable retail deposits run off at 10% or more. The LCR must be at least 100%. No report read gives the rate. | Rule | BCVW-Q10,LCR-Q1,LCR-Q9,LCR-Q10,LCR-D1, MeasurementMap.lcr_ iff,MeasurementMap. control_lcr_needs_ positive_runoff |
| G6 | $a_3$ | The liquidity haircut, the haircut on the risky asset in high-quality liquid assets, the slope $1/a_3$ of the liquidity branch. | The riskless asset counts in full, as Level 1 assets do. The standard's haircuts are 15% (Level 2A) and 25% or 50% (Level 2B). BCVW's baseline 0.30 is none of them. The 40% cap on Level 2 assets and the 75% cap on inflows are not in the model. | Calibration | BCVW-Q11,BCVW-Q12, BCVW-Q13,LCR-Q2,LCR-Q3,LCR-Q4,LCR-Q5,LCR-Q6,LCR-Q7,LCR-Q8,LCR-D2,LCR-D3, MeasurementMap.hqla_ haircut, MeasurementMap.cap_ iff |
| G7 | $q$ | The requirement on $x-1$, with $z=0$ at $x=1+q$. | A floor $\ell_0$ on the leverage ratio is the floor $\ell_0/(1-\ell_0)$ on $x-1$, so the 3% Tier 1 leverage ratio gives $q=3/97$ when Tier 1 equals book equity. This is a reading, and the bundle does not fix $q$. Buffer variance does not depend on $q$. | Rule | B3-Q7,RCR-Q4, MeasurementMap. leverage_req_iff, MeasurementMap. control_leverage_ needs_l0_lt_one, MeasurementMap.zvol2_ free_of_q |
| G8 | $\sigma$, $\mu_s$, $r$ | The volatility and drift of the risky asset and the riskless rate. | No regulatory counterpart. BCVW's baseline is $\sigma=0.08$, $\mu=0.04$, $r=0.02$. | Calibration | BCVW-Q14 |
| G9 | $\sigma_L$, $c$, $\gamma$, $\mu_L$, $r_L$ | The liability shock, its correlation with the risky return, the net income on liabilities, the liability drift and the deposit rate, with $\mu_L=\gamma+r_L$. | No regulatory counterpart. BCVW's baseline is $\sigma_L=0.03$, $c=0.20$, $\gamma=0.02$, $\mu_L=0.03$. | Calibration | BCVW-Q14,M9 |
| G10 | $\rho_L$ | The liability-net discount rate of the reduced problem. | Formed from the shareholder discount rate, BCVW's baseline $\rho=0.12$. No regulatory counterpart. | Calibration | BCVW-Q14 |
| G11 | $\kappa$ | The proportional issuance cost. | No regulatory counterpart. BCVW's baseline is $\kappa=0.01$ for dilution and $\kappa'=0.02$ per unit issued. | Calibration | BCVW-Q14 |
| G12 | $K$ | The fixed issuance cost, charged per unit of liabilities. | Not observed. A cost fixed in dollars would break the reduction to $x$. Locally identified together with $\lambda_S$ from the barrier and the trigger. | Unobserved | BCVW-Q4,K2,M5,M6 |

| ID | Object | Role in the model | Counterpart and observation | Status | Evidence |
|---|---|---|---|---|---|
| G13 | $\lambda_S$ | The asymmetry parameter of the shortfall penalty. | Not observed. The variance ratio carries no information about it. Locally identified together with $K$ from the barrier and the trigger. | Unobserved | `K1,K2,M3,M6,M11` |
| G14 | $y^*$ | The dividend barrier. | The state at which the bank pays dividends (`EQCDIV`). Under the conservation buffer a bank with CET1 between 4.5% and 7.0% of risk-weighted assets has its distributions restricted, so observed payouts reflect the rule as well as the barrier. | Output | `FDIC-Q8,M4,B3-Q4,B3-Q5,B3-Q6,B3-D3` |
| G15 | $x_L$, $y_{\text{post}}$ | The recapitalisation trigger and the injection target. | The state just before and just after a recapitalisation, measured as net stock issuance (`EQCSXQ`) plus capital from the parent holding company (`EQCBHCTR`), in quarters without merger-driven equity changes (`EQCMRG`). | Output | `FDIC-Q9,FDIC-Q10, FDIC-Q12,FDIC-Q13,M5, M6,M7,M10` |
| G16 | $\zeta$, $v_1^2$, $v_3^2$ | Buffer variance, with its deficit limit and its surplus limit. | The variance of changes in $z = \log((x-1)/q)$ near the distress boundary and deep in surplus. It does not depend on $q$. | Output | `M2,M3,M8, MeasurementMap.zvol2_free_of_q` |
| G17 | $v_1^2/v_3^2$ | The deficit-to-surplus variance ratio, read in discrete time as $\lambda_V^4$. | The ratio of the two variances of G16. Reading it as $\lambda_V^4$ is a definition. | Output | `K1,M1,M3` |
| G18 | $z_t$ | The discrete-time state, the log of equity over required capital $Q_t$. | Tier 1 capital over 8.5% of risk-weighted assets, the Tier 1 minimum plus the conservation buffer. | Report | `FDIC-Q3,FDIC-Q4,B3-Q2,B3-Q4,M12` |
| G19 | $g$ | The growth rate of required capital, exogenous in the model. | Quarterly growth of risk-weighted assets. | Report | `FDIC-Q3,M12` |
| G20 | $b$ | The retention ratio. | One minus dividends over net income, over four quarters, defined when net income is positive. | Report | `FDIC-Q7,FDIC-Q8,M12` |
| G21 | $\delta_p$ | The surplus payout rate. | The share of the log-surplus paid as dividends in a quarter. | Report | `FDIC-Q8,M12` |
| G22 | $\phi^-$, $\phi^+$ | Deficit and surplus persistence of $z_t$. | Least squares of $z_t$ on $z_{t-1}$ through the origin, on pairs starting in deficit or surplus. | Output | `M12` |

# 4 Entities

The ledger of the model (`00_reader/ledger_model.py`) is a graph. Its nodes are these entities, the agents and the objects they act on, each observed as stated. Status has the meaning of the Map.

Table 3: Entities of the ledger.

| ID | Entity | Role in the model | Counterpart and observation | Status | Evidence |
|---|---|---|---|---|---|
| N1 | bank | The agent whose policy the model solves. Its state is G1 and its risky exposure G2. | The reporting institution, identified by `CERT`, with total assets `ASSET`, total liabilities `LIAB` and risk-weighted assets `RWAJT`. | Report | `FDIC-Q14,FDIC-Q1, FDIC-Q2,FDIC-Q3` |

| ID | Entity | Role in the model | Counterpart and observation | Status | Evidence |
|---|---|---|---|---|---|
| N2 | equity | The object that dividends draw down and recapitalisations add to (G3). | Tier 1 capital RBCT1J and tier 2 capital RBCT2, net of regulatory adjustments, so not book equity (B3-D2). | Report | FDIC-Q4,RCR-Q1,RCR-Q3,B3-Q3 |
| N3 | investors | Those who receive dividends and supply recapitalisations (G14, G15). | The parent holding company, identified by RSSDHCR, through its transactions EQCBHCTR; outside buyers only in aggregate, through net stock sales EQCSXQ. Whether an investor is the parent holding company is read from RSSDHCR. | Report | FDIC-Q16,FDIC-Q17, FDIC-Q12,FDIC-Q9 |
| N4 | insured creditors | Holders of part of the liabilities, whose shock the model takes as given as one block (G9). | Estimated insured deposits DEPINS, in aggregate per bank; the FDIC's core deposits exclude large time deposits and brokered deposits. | Report | FDIC-Q20,FDIC-Q21 |
| N5 | uninsured creditors | Holders of the rest of the liabilities, which the model does not separate from the insured part (G9). For banks outside the liquidity rule's scope, the only enforcers of the liquidity branch (G5). | Volatile liabilities VOLIAB: federal funds purchased and repurchase agreements, short-term borrowed money, time deposits over $250,000 and foreign office deposits; estimated uninsured deposits DEPUNA, reported only by banks of $1 billion or more. | Report | FDIC-Q22,FDIC-Q23, FDIC-Q24,FDIC-Q25, OCCL-Q1,OCCL-Q2,OCCL-Q3 |
| N6 | supervisor | Enforces the solvency branch of the cap and the requirement (G4, G7), to resolve banks' problems at the least long-term loss to the deposit insurance fund. It enforces the liquidity branch (G5) only for the largest banks. | The bank's primary federal regulator REGAGNT, one of three federal regulators; for a national bank, the Office of the Comptroller of the Currency. | Report | FDIC-Q18,FDIC-Q19, PCA-Q1,PCA-Q2,OCCL-Q1 |
| N7 | insurer | Not in the model as stated. Bears the loss when a bank fails, so it takes part in prompt corrective action and prices its insurance on the bank's capital. | The FDIC, as insurer of the deposits counted in DEPINS. | Report | FDIC-Q20,PCA-Q1,PCA-Q2,ASMT-Q1 |

# 5 Actions

The edges of the ledger: what an agent does to an entity, each leaving a record. A rule enters only through the agent who enforces it; the enforcing actions of the supervisor and the insurer are A9 to A11, from the statute and the regulations read for them.

Table 4: Actions of the ledger.

| ID | Agent, action, entity | Role in the model | Record | Status | Evidence |
|---|---|---|---|---|---|
| A1 | bank pays dividends to investors | The singular control, which draws down equity (G14). A capital distribution under prompt corrective action, barred when it would leave the bank undercapitalized (A9). | Cash dividends on common and preferred stock, EQCDIV, year to date. | Report | FDIC-Q8,PCA-Q3, PCA-Q5 |
| A2 | investors buy stock in equity | One source of a recapitalisation (G15). | Net sale, conversion, acquisition or retirement of capital stock, EQCSXQ, quarterly. It nets the bank's own buybacks and its employee stock ownership plan, which the record does not separate. | Report | FDIC-Q9,FDIC-Q10 |
| A3 | investors inject capital into equity | The other source of a recapitalisation, by the parent holding company (G15). | Transactions with the parent holding company, EQCBHCTR, year to date. They include dividends paid to the parent in property. The supervisor requires a holding company to serve as a source of financial strength for its bank. | Report | FDIC-Q12,PCA-Q9 |
| A4 | insured creditors deposit at and withdraw from the bank | The liability shock, taken as given (G9). | The change in estimated insured deposits DEPINS between reports. | Report | FDIC-Q20 |
| A5 | bank takes risky exposure | The risky exposure, the regular control, bounded by the regulatory cap (G2). | Risk-weighted assets RWAJT over total assets ASSET. | Report | FDIC-Q3,RCR-Q5 |
| A6 | supervisor supervises the bank | Holds the bank to the regulatory cap and the requirement. | The bank's primary federal regulator, REGAGNT. | Report | FDIC-Q18,FDIC-Q19 |
| A7 | uninsured creditors lend to and withdraw from the bank | Not in the model as stated; MMP's wholesale funding. | The change in volatile liabilities VOLIAB between reports, and in estimated uninsured deposits DEPUNA for banks of $1 billion or more. | Report | FDIC-Q22,FDIC-Q23, FDIC-Q24,FDIC-Q25 |
| A8 | insurer insures the insured creditors | Not in the model as stated. | The estimated insured deposits DEPINS. | Report | FDIC-Q20 |
| A9 | supervisor assigns the bank a capital category, with its restrictions | Enforces the requirement: the bank's category sets which actions bind, from the bar on distributions (A1) to an order to sell shares. | The capital category follows from the ratios the bank reports: well capitalized needs a total risk-based ratio of at least 10.0%, adequately capitalized at least 8.0%. Read from Tier 1 capital RBCT1J, tier 2 capital RBCT2 and risk-weighted assets RWAJT; the directives themselves are not public. | Rule | OCC-Q1,OCC-Q2,OCC-Q3,OCC-Q4,OCC-Q6, PCA-Q5,PCA-Q6,PCA-Q7,RCR-Q7 |

| ID | Agent, action, entity | Role in the model | Record | Status | Evidence |
|---|---|---|---|---|---|
| A10 | supervisor closes the bank | The counterpart of the model's distress boundary (G1 at one, equity zero), reached earlier: within 90 days of a bank's tangible equity falling to 2% of total assets, the supervisor must appoint a receiver or take another action the insurer concurs in. | Receivership after critical undercapitalization; the FDIC's record of failed banks is not among the sources read. | Rule | PCA-Q4,PCA-Q8,OCC-Q5 |
| A11 | insurer prices the bank's deposit insurance | Not in the model as stated. A bank with more capital pays less for its insurance. | The assessment rate of a small bank, the uniform amount plus pricing multipliers times seven financial ratios, the first the leverage ratio; the multiplier on the leverage ratio is negative (ASMT-D1). | Rule | ASMT-Q1,ASMT-Q2, ASMT-Q3,ASMT-Q4 |

# 6 What the empirical analysis measured

Each object of the map, and what the FDIC panel of US banks, 2015 to 2024, could do with it. Tracked means the object is measured and no model prediction about it is tested. Holds and Fails mean a prediction of the solved model about it is tested. Not measured means the data available do not measure the object. Each figure is printed by the computation in the cited results file, which the checker re-runs on its pinned data.

Table 5: What the data could measure.

| ID | Measured as | What the data show | Model prediction | Outcome | Evidence |
|---|---|---|---|---|---|
| G1 | ASSET/LIAB, 213,771 bank-quarters. | Median 1.1182, interquartile range 1.0997 to 1.1447. | None on its own. | Tracked | FDIC-Q1,FDIC-Q2, 06_empirical_ starter/results_ model_and_data.md |
| G2 | RWAJT/ASSET, under the risk-weight identification. | Median 0.6857. | $\pi = \bar{\pi}$, 2.3794 at the median state. | Fails | FDIC-Q3, MeasurementMap.rwa_ two_assets,06_ empirical_starter/ results_model_and_ data.md |
| G3 | Book equity over assets, and Tier 1 capital (RBCT1J). | Book equity over assets has median 10.57%, interquartile range 9.06% to 12.64%. | None on its own. | Tracked | FDIC-Q4,06_ empirical_starter/ results_model_and_ data.md |
| G4 | The bank's Tier 1 capital over risk-weighted assets. | Median 15.04%, 3.34 times $a_1$, and 2.78 times at the 25th percentile. | The ratio sits at $a_1$. | Fails | FDIC-Q4,B3-Q1,B3-D1,06_empirical_ starter/results_ model_and_data.md |
| G5 | Not reported in the data read. | Nothing. | None tested. | Not measured | G5 |
| G6 | Not reported in the data read. | Nothing. | None tested. | Not measured | G6 |
| G7 | Not reported. The leverage ratio gives a candidate floor. | Nothing. | None tested. | Not measured | G7 |

| ID | Measured as | What the data show | Model prediction | Outcome | Evidence |
|---|---|---|---|---|---|
| G8 | The volatility of book $x$ is computed, which is not the economic $\sigma$. | Book volatility of $x$ 0.01468 a year. The model's volatility of $x$ is 0.06081 at the observed $\pi$, 4.1 times larger. | A level, on an economic-value basis. | Not measured | `06_empirical_starter/results_model_and_data.md` |
| G9 | Not separately measured. | Nothing. | None tested. | Not measured | `G9` |
| G10 | Not measured. | Nothing. | None tested. | Not measured | `G10` |
| G11 | Not measured. | Nothing. | None tested. | Not measured | `G11` |
| G12 | Through the trigger and target of G15, which the data do not show. | No trigger-and-target behaviour (G15). | Identified with $\lambda_S$ from the thresholds. | Not measured | `K2,06_empirical_starter/results_distress_targeting.md` |
| G13 | As G12. | As G12. | As G12. | Not measured | `K2,06_empirical_starter/results_distress_targeting.md` |
| G14 | Quarters with a dividend (`EQCDIV`), by $x$. | 56.2% of bank-quarters below $y^* = 1.5715$ pay a dividend. Payment falls as $x$ rises, from 60.1% for $x$ between 1.03 and 1.10 to 32.0% between 1.20 and 1.39. | No dividend below $y^*$. | Fails | `FDIC-Q8,M4,06_empirical_starter/results_dividend_location.md` |
| G15 | Recapitalisations, $x$ before and after, within bank, under the pre-registered rule. | Issue size is unrelated to the pre-issue state and the post-issue state rises with it, $b_{\text{post}} = +1.37$ and $+1.36$ for the two valid distress proxies. Issues start at a mean $x$ of 1.11, against the trigger 1.03. | One target within a bank, $b_{\text{post}} = 0$. | Fails | `FDIC-Q9,FDIC-Q12,M5,06_empirical_starter/PREREG_distress_targeting.md,06_empirical_starter/results_distress_targeting.md` |
| G16 | Not computed. A level would need the economic $\sigma$ of G8. | Nothing. | $v_1^2, v_3^2$, which need the cap to bind (M3), and it does not (G2). | Not measured | `M3` |
| G17 | Not computed on this panel. | Nothing. | $v_1^2/v_3^2$, under the same condition as G16. | Not measured | `K1,M3` |
| G18 | Tier 1 capital over 8.5% of risk-weighted assets. | 462 of 130,488 quarter pairs start in deficit, about 0.35%. | None on its own. | Tracked | `FDIC-Q3,FDIC-Q4,06_empirical_starter/results_persistence.md` |
| G19 | Quarterly growth of risk-weighted assets. | Mean $-0.47\%$ a quarter on deficit pairs: banks in deficit shrink risk-weighted assets. | Exogenous, so no response to a deficit. | Fails | `FDIC-Q3,M12,06_empirical_starter/results_persistence.md` |
| G20 | One minus dividends over net income, four quarters. | 0.3131 on the 108 deficit pairs where it is defined in $(0,1]$. | None on its own. | Tracked | `FDIC-Q7,FDIC-Q8,06_empirical_starter/results_persistence.md` |
| G21 | Share of the log-surplus paid out per quarter. | 0.0211 a quarter. | None on its own. | Tracked | `FDIC-Q8,06_empirical_starter/results_persistence.md` |

| ID | Measured as | What the data show | Model prediction | Outcome | Evidence |
|---|---|---|---|---|---|
| G22 | Within the pre-registered tests T1 to T3. | Deficit persistence 0.88 against 1.01 predicted (T1, fails). Surplus persistence 0.9959 against 0.9789 (T3, fails). The fall of $\phi^-$ in $g$ is not measured (T2). | $\phi^- = 1-g(1-b)/b$, $\phi^+ = 1-\delta_p$. | Fails | `M12,06_empirical_starter/PREREG_persistence.md,06_empirical_starter/results_persistence.md` |

# 7 The ratio constraints in the state

With equity $X-L$, risk-weighted assets $\pi X$, high-quality liquid assets $X-a_3\pi X$ and net outflows $a_2 L$, the two ratio constraints of BCVW are the two branches of the cap in $x = X/L$. For $a_1, a_2, a_3, \pi, X, L > 0$,

$$a_1 \le \frac{X-L}{\pi X} \iff \pi x \le \frac{x-1}{a_1}, \qquad 1 \le \frac{X-a_3\pi X}{a_2 L} \iff \pi x \le \frac{x-a_2}{a_3},$$

and both hold if and only if $\pi x \le u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ (`MeasurementMap.solvency_iff`, `MeasurementMap.lcr_iff`, `MeasurementMap.cap_iff`). At $\pi = 0$ the left side of the first equivalence is undefined and the equivalence fails, and at $a_2 = 0$ the same holds for the second (the two controls). For $\ell_0 < 1$ and $X, L > 0$, $\ell_0 \le (X-L)/X \iff \ell_0/(1-\ell_0) \le x-1$ (`MeasurementMap.leverage_req_iff`).

# 8 Pending

- The CET1 ratio (`RBCT1CER`) is not yet in the panel. The corner test of G4 uses Tier 1 against the CET1 minimum (B3-D1).
- The unit of the FDIC amounts, thousands of dollars, is read from one API record and not from the workbook (FDIC-D1).
- Rows G1 to G3 assume regulatory capital equals book equity. Items 12 to 18 and 27 to 29 of Schedule RC-R, which give the adjustments, were not read.
- The tier that $a_1$ stands for is not fixed (B3-D1).

# References


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